

Math 421

Wednesday, October 1

Announcements

- Homework 4 due Friday
- Grace's Drop-In Hours
 - **Today:** 12-1 VV 311
 - Thursday: 9-11 VV B205
- MLC
 - Proof Table: M-Th 3-7:30 VV B227
 - Course Assistant: Th 4-7 Table 4
- Exam 1 – 10/15

Example

Theorem. If $\lim s_n = s$, $s_n \geq 0$ for all n , and $s > 0$, then $\lim \sqrt{s_n} = \sqrt{s}$.

Recall – Discussion. $|\sqrt{s_n} - \sqrt{s}| < \varepsilon \iff \underbrace{|s_n - s|} < \sqrt{s} \varepsilon$

Proof. Let $\varepsilon > 0$. Since $\lim s_n = s$, there exists some N such that if $n > N$, then $|s_n - s| < \sqrt{s} \varepsilon$. Assume $n > N$, then

$$|\sqrt{s_n} - \sqrt{s}| = \left| (\sqrt{s_n} - \sqrt{s}) \cdot \frac{\sqrt{s_n} + \sqrt{s}}{\sqrt{s_n} + \sqrt{s}} \right| = \left| \frac{s_n - s}{\sqrt{s_n} + \sqrt{s}} \right| = \frac{|s_n - s|}{\sqrt{s_n} + \sqrt{s}}$$

bounded the expression involving $\sqrt{s_n}$ to remove dependence on n

$$\leq \frac{|s_n - s|}{\sqrt{s}} < \frac{\sqrt{s} \varepsilon}{\sqrt{s}} = \varepsilon.$$

to introduce a $|s_n - s|$ term.

Since $\varepsilon > 0$ was arbitrary, $\lim \sqrt{s_n} = \sqrt{s}$.

Scratch work goal:
"solve" for $|s_n - s| < \dots$

cannot involve n .

Activity – Fill in the Blanks

Theorem. If $\lim s_n = s$, then $\lim 5s_n = 5s$.

Proof. Fix $\epsilon > 0$. Since $\lim s_n = s$, there exists some N such that: if $n > N$, then $|s_n - s| < \underline{\epsilon/5}$.

$\frac{\epsilon}{|k|}$ must be ϵ

Assume $n > N$. Then

$$|5s_n - 5s| = |5| |s_n - s| = 5 |s_n - s| < 5 \cdot \frac{1}{5} \cdot \epsilon = \epsilon.$$

Since $\epsilon > 0$ was arbitrary, $\lim 5s_n = 5s$.

Q: What would need to be modified if you wanted to instead to prove

$\lim ks_n = ks$, for $k \in \mathbb{R}$?

$\wedge k \neq 0$

$$|s_n^2 - s^2| = |(s_n + s)(s_n - s)| = \underbrace{|s_n + s|}_{\leq M + |s|} |s_n - s| < \epsilon$$

Activity

$$|s_n + s| \leq |s_n| + |s| \leq M + |s| \leq (M + |s|) |s_n - s| < \epsilon$$

Theorem. If $\lim s_n = s$ and $s_n \in [-M, M]$ for all n , then $\lim(s_n)^2 = s^2$.

$$|s_n - s| < \frac{\epsilon}{M + |s|}.$$

1. **Scratch Work:** Find an upper bound for $|s_n^2 - s^2|$ in terms of $|s_n - s|$.

- **Hint:** $a^2 - b^2 = (a + b)(a - b)$.
- **Reminder:** The only term in the bound that can depend on n is $|s_n - s|$ itself. Everything else must be independent of n .

2. **Prove Theorem.**

Activity

Theorem. If $\lim s_n = s$ and $s_n \in [-M, M]$ for all n , then $\lim(s_n)^2 = s^2$.

Proof. Let $\varepsilon > 0$. Since $\lim s_n = s$, there exists some N such that if $n > N$, then $|s_n - s| < \frac{\varepsilon}{M + |s|}$. Assume $n > N$. Then

$$|s_n^2 - s^2| = |s_n + s| |s_n - s| \leq (M + |s|) |s_n - s| < M + |s| \cdot \frac{\varepsilon}{M + |s|} = \varepsilon.$$

Since $\varepsilon > 0$ was arbitrary, $\lim s_n^2 = s^2$.